

Emergent Spacetime and Topological Matter Fields from Coupled Chaotic Networks: The Dot-Dynamics Paradigm

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Abstract—This paper presents a discrete computational model for the unification of spacetime topology and matter fields. By abandoning the assumption of a continuous background metric, we demonstrate that a lattice network of coupled non-linear chaotic kernels naturally undergoes period-doubling bifurcation to generate stable, conserved topological defects. In the continuum limit, these defects map directly to the stress-energy tensor within Einstein’s Field Equations, offering an algorithmic bridge between discrete information networks and continuous general relativity.

I. INTRODUCTION

Modern physics remains divided between the continuous geometry of General Relativity and the discrete mechanics of Quantum Field Theory. We propose a paradigm shift based on the foundational framework: $0 \rightarrow 10$ (Binary) $\rightarrow \infty$.

II. THE CORE FORMULATION

The evolution of the localized system state Ψ_i at discrete time step $n + 1$ across a topological ring network is defined as:

$$\Psi_i(n + 1) = (1 - \alpha)f(\Psi_i(n)) + \alpha \sum_j W_{ij}f(\Psi_j(n)) \quad (1)$$

Where $f(\Psi) = r\Psi(1 - \Psi)$ serves as the localized chaotic kernel engine.

III. COMPUTATIONAL VERIFICATION

Using an optimized array processing matrix, numerical simulations demonstrate that when the driving parameter $r > 3.57$, the system transcends rigid crystalline lattice states (10) and self-organizes into fluidly woven spatiotemporal structures (∞).

IV. CONCLUSION

The spatial and temporal gradients of these emergent network variations explicitly construct a macroscopic Stress-Energy Tensor $T_{\mu\nu}$, directly sourcing the curvature of the spacetime metric $g_{\mu\nu}$.